Synopsis

Project Title: Predictive Analytics for Device Usage and Thermal Health

Problem Statement 24: Design and develop an end-to-end machine learning classification system to analyse smartphone sensor usage data and predict high-usage or overheating conditions. The project involves selecting an appropriate smartphone sensor dataset containing features such as screen time, motion sensor readings, touch event frequency, battery consumption, and device activity levels. Data preprocessing steps including data cleaning, handling missing values, normalisation, and feature transformation are applied to ensure data quality and consistency. Exploratory Data Analysis is performed to study feature distributions, identify dominant usage factors, analyse relationships between sensor activity and battery consumption, and detect outliers or abnormal usage patterns using statistical summaries and visualisations. Correlation analysis is conducted to identify key indicators of heavy smartphone usage. Based on insights obtained from EDA, relevant features are selected to improve model interpretability and performance (PCA, Forward Selection, Backward Selection). Machine learning classification models are then developed and evaluated to classify smartphone usage as normal or heavy and to detect possible overheating scenarios, demonstrating a complete machine learning workflow from raw data to evaluated predictive models.

Team ID: PY-GT-14

Student Names:

1.Parikshit Thakur [TEAM LEADER]

University Roll number:

2028460

Section: G

2.Aditya Verma [TEAM MEMBER]

University Roll number:

2028384

Section: G

3.Siddhant Tiwari [TEAM MEMBER]

University Roll number

2028499

Section:G

Problem Description:

Smartphones have become an essential part of daily life, but excessive usage often leads to problems such as **rapid battery drain, device overheating,** and **reduced performance.** Many users are unaware of how their usage patterns—like long screen time, frequent touch interactions, and continuous background activity—affect their device's health. The main challenge is to analyse large amounts of smartphone sensor data and identify patterns that indicate high usage or potential overheating conditions. This data, collected from various sensors such as motion sensors, battery monitors is often complex and unstructured, making it difficult to interpret manually.

Therefore, there is a need to develop an intelligent system that can automatically process and analyse this data to detect abnormal or heavy usage behaviour. By applying machine learning techniques, the system can classify usage patterns and provide early warnings about possible overheating risks. This will help users manage their smartphone usage more effectively.

Steps of implementation:

1. Data Collection:

We collect data from phone sensors like screen time, touch activity, and battery usage. This data gives a clear picture of how the phone is being used.

2. Data Preprocessing:

Raw data is often messy, so we clean it and fix missing values. We also scale and transform it so the machine learning model can understand it properly.

3. Model Selection:

Different classification models like **Logistic Regression, Decision Tree,** and **Random Forest** are tested on the data.

4.Training and Validation:

During **training,** the selected model learns patterns from the smartphone sensor data like usage and battery behaviour.In **validation,** the model is tested on unseen data to check if it can accurately predict high usage and overheating conditions.

5.Evaluation:

In evaluation, we check how well the model predicts smartphone usage and overheating using metrics like accuracy and F1-score.

6.Final Model:

The final model is the one that performs best after comparing all the trained models.
It is used to accurately predict high smartphone usage and possible overheating in real-world scenarios.

Expected Modules/Libraries to be used:

- **NumPy** – for handling numerical data and calculations.
- **Pandas** – for loading, cleaning, and managing the dataset.
- **Matplotlib / Seaborn** – for creating graphs and visualising data patterns.
- **Scikit-learn (sklearn)** – for preprocessing, feature selection, and machine learning models.
- **SciPy** – for statistical analysis.

Expected Evaluation Metrics and Plots:

In this project, we will use metrics like **accuracy, precision, recall, and F1-score** to check how well our model is performing. These help us understand not just if the model is correct, but also how well it detects important cases like heavy usage or overheating.For better understanding, we will use visualisations like **heatmaps, histograms, and box plots.** These graphs make it easier to see patterns in the data, spot mistakes made by the model, and identify unusual smartphone usage behaviour.

Required topics from the subject:

Files

Files are used to store and read the smartphone sensor dataset.
They help us load data into the program and save results like predictions or reports.

Strings

Strings help in handling text data like labels (e.g., "normal" or "high usage").
They make it easy to manage and process non-numerical information in the dataset.

Arrays

Arrays are used to store large amounts of sensor data like screen time and battery values.
They help in performing calculations and passing data to machine learning models efficiently.

Functions:

Functions help in organising the code into smaller, reusable parts.

Platform Required:

Jupyter Notebook.

Code Blocker.

Visual Studio / Python IDLE.

Google Collab.

Books and Link Sources:

Books:

1.*Hands-On Machine Learning with Scikit-Learn,*

Dataset Link: Embedded Smartphone Sensor Data (Kaggle)

UCI Machine Learning Repository: WISDM Dataset

https://share.google/cTWIq1itddydc1P4U

Sample Dataset:

A sample smartphone sensor usage dataset with 15 rows is included below for reference. The target column indicates whether the device is experiencing normal usage, high usage, or potential overheating conditions.

User_ ID	Screen _Time (hrs/ day)	Motion_ Activity (avg/ min)	Touch_ Events (per min)	Battery _Usage (%/hr)	CPU _Usa ge (%)	App_Usa ge_Coun t	Temp eratu re (°C)	Idle _Ti me (hrs)	Usage_Label
U001	2.5	15	120	5	30	12	34	10	Normal
U002	6.8	40	300	12	65	25	41	4	High Usage
U003	8.2	55	450	18	85	35	47	2	Overheating
U004	1.8	10	90	4	25	8	32	12	Normal
U005	5.5	35	260	10	60	20	39	5	High Usage
U006	9.0	60	500	20	90	40	50	1	Overheating
U007	3.2	20	150	6	40	15	35	8	Normal
U008	7.5	50	380	14	75	30	43	3	High Usage
U009	10.0	70	550	22	95	45	52	1	Overheating
U010	4.0	25	200	8	50	18	37	6	Normal
U011	6.0	38	280	11	62	22	40	4.5	High Usage
U012	2.0	12	100	5	28	10	33	11	Normal
U013	8.8	65	480	19	88	38	48	2	Overheating
U014	5.0	30	230	9	55	19	38	6	High Usage
U015	3.5	18	160	7	42	14	36	7	Normal